

Reproduction quickstart

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Every number in the paper comes from a JSON certificate in this repository. Nothing is typed by hand, and the build fails if an artifact is missing.

Setup

```
git clone <repository>
cd selfdual-5form-invariants
python -m venv .venv && .venv/bin/pip install -r requirements-lock.txt
Python 3.13, NumPy 2.5.1, pynauty 2.8.8.1. No other dependency.
```

The five checks worth running first

```
# 1. the orientation fix, independently verified, ~10 min
.venv/bin/python scripts/verify_orientation_canonical_independent.py

# 2. dim_Q D10 = 11 by a route sharing no code with production, ~3 min
.venv/bin/python scripts/verify_D10_independent.py

# 3. dim_Q Q10 = 3, constructed rather than inferred, ~3 min
.venv/bin/python scripts/construct_Q10_exact.py

# 4. the assembled rank matrix, read-only, seconds
.venv/bin/python spinor_trace_bridge/scripts/assemble_rank81_matrix.py

# 5. the science gate over every artifact, seconds
.venv/bin/python scripts/emit_jhep_science_gate.py

Expected: PROVED, PROVED, MATRIX COMPLETE, VERDICT: PASS.
```

Tests

```
.venv/bin/python -m pytest tests/ -q          # tensor side
cd spinor_trace_bridge && python -m pytest -q  # bridge side
```

Rebuilding the paper

```
.venv/bin/python manuscript/jhep/make_jhep_assets.py
```

```
cd manuscript/jhep && tectonic -X compile main.tex --outdir .
```

The asset script regenerates every figure, table and inline number from the certificates first, so a stale value cannot survive a rebuild.

The expensive one

Recomputing the full certificate matrix from scratch takes several hours on one laptop: eighteen cells, one at a time, and the cold ones run 20 to 35 minutes each.

```
sh spinor_trace_bridge/scripts/run_rank81_matrix.sh <archive> <python>
```

Warm row caches make repeat runs seconds per cell. The archive argument is a third-party candidate-selection list that is not redistributed; every other result reproduces without it.

If a number disagrees

The certificate is authoritative, not the paper. Each claim in the ledger names the artifact it came from.